Challenges in Machine Learning - Lecture 3 Summary

This lecture covers four central challenges in Machine Learning: **1. Overfitting**
Models that perform very well on training data but fail to generalize. Often caused by noisy or non-representative data. Example: deep decision trees or high-degree polynomial regression. **2. Cross-Validation**
A method to estimate model performance and avoid overfitting. K-fold cross-validation is widely used, while in data-rich cases we split into training, validation, and test sets. **3. The Curse of Dimensionality**
As dimensionality increases, problems that are easy in low dimensions become hard. More data is required, and distance-based methods struggle. Solutions include dimensionality reduction and feature selection. **4. Bias-Variance Trade-off**
Prediction error decomposes into Bias² + Variance. Low-complexity models suffer high bias; high-complexity models suffer high variance. The key is to balance model complexity with data resources. **Take-home message:** Match model complexity to the data you have, not just the target complexity.

